NORTHSTAR MANUFACTURING UK

Project guide & real UI screenshots

What was built, what runs locally, what is simulated, and how to put the whole project on GitHub.

Handover: 12 October 2026 · Synthetic baseline: 11 October 2026 Independent simulated portfolio · Free-tool route

Built: a read-only interactive portfolio plus all twelve phases of draft analysis artifacts.

Not built: a live SAP implementation or a transactional ERP backend. Learning review and real business acceptance remain open.

Start with the overview, then follow the local and GitHub steps. The companion Full Portfolio PDF contains the detailed catalogues; editable sources and raw data stay in the repository.

What was built — NorthStar project guide

Independent simulated portfolio. Initial build 11 October 2026; handover refreshed 12 October 2026. This guide describes actual artifacts and clearly separates the running portfolio application from a future SAP implementation. The synthetic data snapshot remains 11 October 2026.

1. The direct answer

Yes: the twelve-phase **artifact pack** and a **read-only, interactive portfolio application** have been built. You can run the application locally, inspect records and analysis, read the documents, and publish the static application using GitHub Pages after uploading the repository and enabling Pages.

No: this is not a working SAP ERP implementation or a transactional order-management backend. Creating orders, reserving stock, enforcing credit approvals, posting goods issue, issuing legally valid invoices and clearing real payments are requirements/design scenarios in the portfolio. They are not executable business transactions in this app.

The earlier phrase “full project built” referred to the artifact pack and viewer. It did not mean a production ERP system, completed SAP UAT or verified learner mastery. Those distinctions are essential to honest portfolio use.

2. What you can actually do in the application

Screen	Implemented behaviour	Boundary
Overview	Six calculated synthetic KPI cards, exclusions and representative order link	Snapshot, not live business performance
Orders	Search by order/customer/issue; status filter; export filtered CSV; open linked delivery/invoice/allocation detail	Read-only records; no order creation or modification
Process maps	Switch between AS-IS and TO-BE SVG flowcharts	Proposed process logic, not a workflow execution engine
Requirements	Search 23 requirements and inspect acceptance, dependencies and trace links	Proposed requirements, not stakeholder-approved scope
Data & KPIs	Inspect a chart, formulas, populations and limitations	Generated snapshot; filters on the order list do not recalculate cards
UAT & checks	Expand 22 test designs with prerequisites, steps and expected results	Every SAP case remains Not executed
SAP evidence	Read nine source-register entries and open official links	Seven reviewed sources, two unverified pages; no tenant demonstration
Document library	Select embedded phase documents and download PDF/workbook/presentation/setup files	Download files require the complete project folder or hosted repository

There are no logins, roles enforced by the app, live SAP APIs, cloud databases, payment services or secret keys. The role-access matrix describes a proposed business control; it is not implemented authentication. The SQLite database is an analytical output for local inspection, not a backend used by the browser.

3. What work was done

- Converted the supplied scope into a charter, representative order, learning plan and maintained registers.

- Created explicitly simulated stakeholder discovery and mapped conflicting needs to proposed policies.

- Built AS-IS and TO-BE flowcharts with ownership, handoffs and exceptions; separated hypotheses from proven causes.

- Authored 3 business, 15 functional and 5 non-functional requirements, 12 business rules, 15 stories and 22 UAT cases; linked them using stable IDs.

- Reviewed official SAP sources and pinned the reference to S/4HANA Cloud Public Edition 2602; recorded access dates and uncertainty.

- Built a reproducible 300-order synthetic dataset with separate customer, product, order, delivery, invoice, receipt, allocation and event tables.

- Implemented six KPI calculations in Python and independently in SQLite; executed 24 data checks and a hand-calculated fixture.

- Built a ten-risk register, role change/training/support plan and illustrative option comparison that separates capacity from cash savings.

- Created an offline HTML/CSS/JavaScript portfolio viewer, editable workbook and report, ten-slide presentation, case study and conditional interview/resume pack.

- Added local serving, root website entry, GitHub Pages preparation, real-browser checks, screenshot evidence and PDF handover materials.

AI performed the drafting, coding and recorded tool execution in this conversation. Your personal contribution so far is supplying the brief, scope, background and preferences. You have not yet demonstrated independent review or reproduction here. Update that statement as you do the practical tasks; do not claim personal implementation/testing work merely because artifacts exist.

4. All twelve phases and their outputs

Phase	What is present	What remains
1 Charter	Scope, objectives, assumptions, decisions, schedule, NS-ORD-1001	Tailoring to actual study time and primary target role
2 Discovery	Stakeholder matrix, questions, role-play, findings and RACI	Real stakeholder engagement
3 AS-IS	Flowchart, eight-step register and root-cause hypotheses	Validate current process and causes with evidence
4 Requirements	BRD, 23 requirements, 12 rules, access and traceability	Business review and policy approval
5 TO-BE/SAP	Future flow, controls, data needs and source register	Tenant-specific feasibility and settings
6 Fit-gap	Fifteen functional desk assessments plus NFR discussion	System demonstrations against criteria
7 Backlog	Fifteen stories with criteria, dependencies and exceptions	Delivery planning and implementation
8 Analysis	300 orders, dictionary/model, six KPIs, SQL, checks and charts	Broader real data and baseline validation
9 UAT	Twenty-two cases, criteria/severity and design walkthrough	SAP execution and user acceptance
10 Change/case	Ten risks, role impacts, training/support and scenario costs	Agreed operating model and business case
11 Packaging	README, viewer, editable files, deck and case study	Optional GitHub publication by you
12 Coaching	Conditional project descriptions, interview scripts and questions	Your answers, feedback and mastery review

5. How the project works technically

The source files are Markdown documents, CSV catalogues, authored Python, JavaScript and CSS. Python generates deterministic synthetic tables and calculations, then builds the viewer by embedding a JSON snapshot and the documents into HTML. That file runs in the browser without a backend. The same static files can be served locally or by GitHub Pages.

Flow: authored catalogues and generator → CSV / SQLite / calculated results → generated HTML and editable reports → browser exploration. No browser action writes to the original CSVs. Editing the workbook alone does not update the viewer. To change its source data or generated documents consistently, modify source inputs and run the build scripts again.

Root index.html, viewer/index.html and downloads/NorthStar_Offline_Portfolio.html are generated copies of the same app. The root copy makes local and GitHub Pages entry straightforward. Source JavaScript and CSS stay in viewer/app.js and viewer/style.css. This intentional duplication supports a portable single-file app as well as an editable repository.

6. Data and KPI interpretation

The dataset contains 300 orders, 285 deliveries, 262 invoices, 209 receipts and 197 allocations. It includes split deliveries, partial payments, open orders and unmatched receipts. Dates are synthetic UK business-local calendar dates with a snapshot of 11 October 2026. Amounts use integer pence; invoices and receipts are net-only and omit VAT. One line per order is a teaching simplification.

The six full-cohort results are: cycle time 6.16 calendar days; on-time delivery 49.07%; quantity fulfilment 84.89%; billing cycle 1.95 calendar days; order error rate 22.33%; manual-touch rate 76.33%. These are properties of generated data, not actual NorthStar results or achieved improvements.

Cycle time includes 232 fully received orders and excludes 51 open noncancelled orders. On-time delivery counts 132 successes among 269 due initial promises, including overdue open failures. One unknown promise is separately disclosed. Billing cycle covers 262 invoiced deliveries and excludes 22 goods-issued unbilled deliveries. Read the KPI contract before interpreting any number.

The model aggregates child records before combining order measures. A delivery × payment join can otherwise multiply values. A partial payment's analytical remaining balance is not proof that an SAP invoice has been cleared.

7. How to run it locally

Extract the repository ZIP, open a terminal in the northstar-otc folder and run `python scripts/serve.py --open`. Visit `http://127.0.0.1:8000/` if it does not open. Windows can use `py`; macOS/Linux may use `python3`. The server needs only Python's standard library and stops with Ctrl+C.

Double-clicking index.html is also an intended offline path for browsers allowing file URLs. This environment's browser policy blocks that protocol, so actual verification used local HTTP. Refer to RUN_LOCALLY.md for alternatives, the five-minute tour, rebuild commands and troubleshooting.

8. GitHub access and the free route

Push the complete folder contents to a new public repository. Keep index.html at the root. Enable Settings → Pages → Deploy from a branch → main → /(root). After deployment, use the actual URL GitHub displays. A typical project URL has the form https://YOUR_USERNAME.github.io/northstar-otc/; this is a format example, not an existing deployed site.

GitHub Pages can host this static project from a public repository on GitHub Free. The one-time Pages setup is needed; pushing alone is insufficient. GITHUB_SETUP.md includes exact commands, GitHub Desktop guidance and official sources accessed on 11 October 2026. No custom domain or paid service is required for this planned route. Existing device, internet and chat/cloud costs are separate.

Nothing has been pushed to your account or deployed by this handover. GitHub hosting must be checked after your own deployment.

9. Verification and screenshot meaning

Python/SQLite data checks and catalogue consistency checks are recorded in quality and phase-08/data. Node smoke checks verify viewer logic. Real Chromium UI checks, when present, are recorded separately in quality/browser-checks.json with the tested HTML hash, browser version and screenshot references. Screenshots in the handover PDF come from that running local portfolio, not the earlier generated concept image and not SAP software.

The browser exercise covers navigation, search, filters, CSV download, record detail, requirement links, UAT expansion, process switching, evidence, document selection, download links, repository-prefix paths and a mobile viewport. It is not a comprehensive security/accessibility certification or real SAP UAT. No real business users were interviewed or tested. LibreOffice visual rendering is not verified.

10. What “complete” means here

The twelve-phase artifact pack and defined read-only viewer are built. Their exact execution checks and remaining limits are reported, rather than calling them production ready. The original final learning gate stays open until you can explain the decisions, calculations and your personal contribution independently. The files do not prove a live SAP implementation or guarantee employment.

Turning this into a transactional ERP application would be a separate software-development scope: persistence, authenticated roles, transaction rules, inventory/reservation logic, accounting semantics, integrations, security, operational support and system acceptance. That work is neither required for this BA portfolio nor silently claimed as completed.

11. Where to start learning

Read Phase 1 and explain why 60 on hand cannot confirm 100 requested. Follow FR-004 through its story and tests. Recalculate the four-order fixture. Explain physical dispatch versus posting versus receipt. Then work through the short task and three questions in each phase. Resume wording should evolve only as you personally review, change, reproduce and explain the work.

12. Handover files

The Project Guide PDF explains the build, real screenshots, local use and GitHub setup. The Full Portfolio PDF supplies the narrative documents and detailed catalogue appendices. Editable originals and raw synthetic data remain in the repository because PDFs are not a suitable replacement for executable scripts or tabular source data. All formats are readable with free tools.

Real browser verification

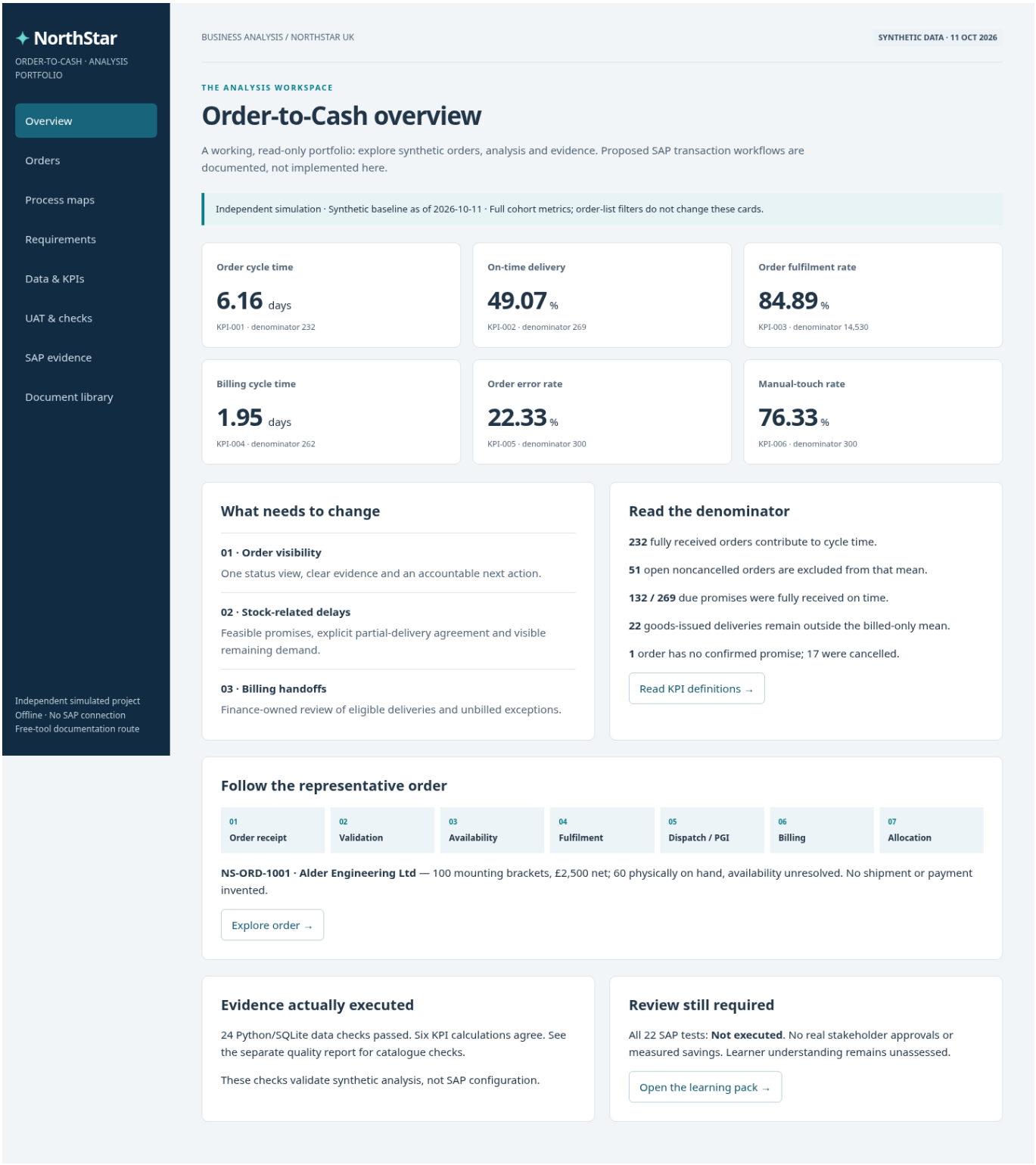
20 real Chromium checks passed. Browser 151.0.7922.173. Execution timestamp (UTC): 2026-10-12T15:56:01.018042+00:00.

Tests exercised the portfolio over HTTP on a temporary loopback server. No GitHub deployment or SAP UAT occurred. Screenshots are direct browser captures; they are not generated concept artwork.

Check	Result
Root entry opens over local HTTP	Passed
Six KPI cards render expected synthetic values	Passed
Desktop overview fits viewport	Passed
Representative order search and detail	Passed
No-result search state	Passed
Status filter and actual CSV download	Passed
Shipment/invoice/allocation drill-down	Passed
Requirement search and traceability	Passed
UAT details and honest statuses	Passed
AS-IS / TO-BE flowchart switching	Passed
Data analysis screen	Passed
Evidence register	Passed
Document library selection	Passed
Library download links resolve	Passed
GitHub-style repository subpath	Passed
Alternative prebuilt entry points	Passed
Keyboard navigation sample	Passed
Mobile layout sample	Passed
JavaScript runtime errors	Passed
Self-contained runtime	Passed

Overview and KPIs

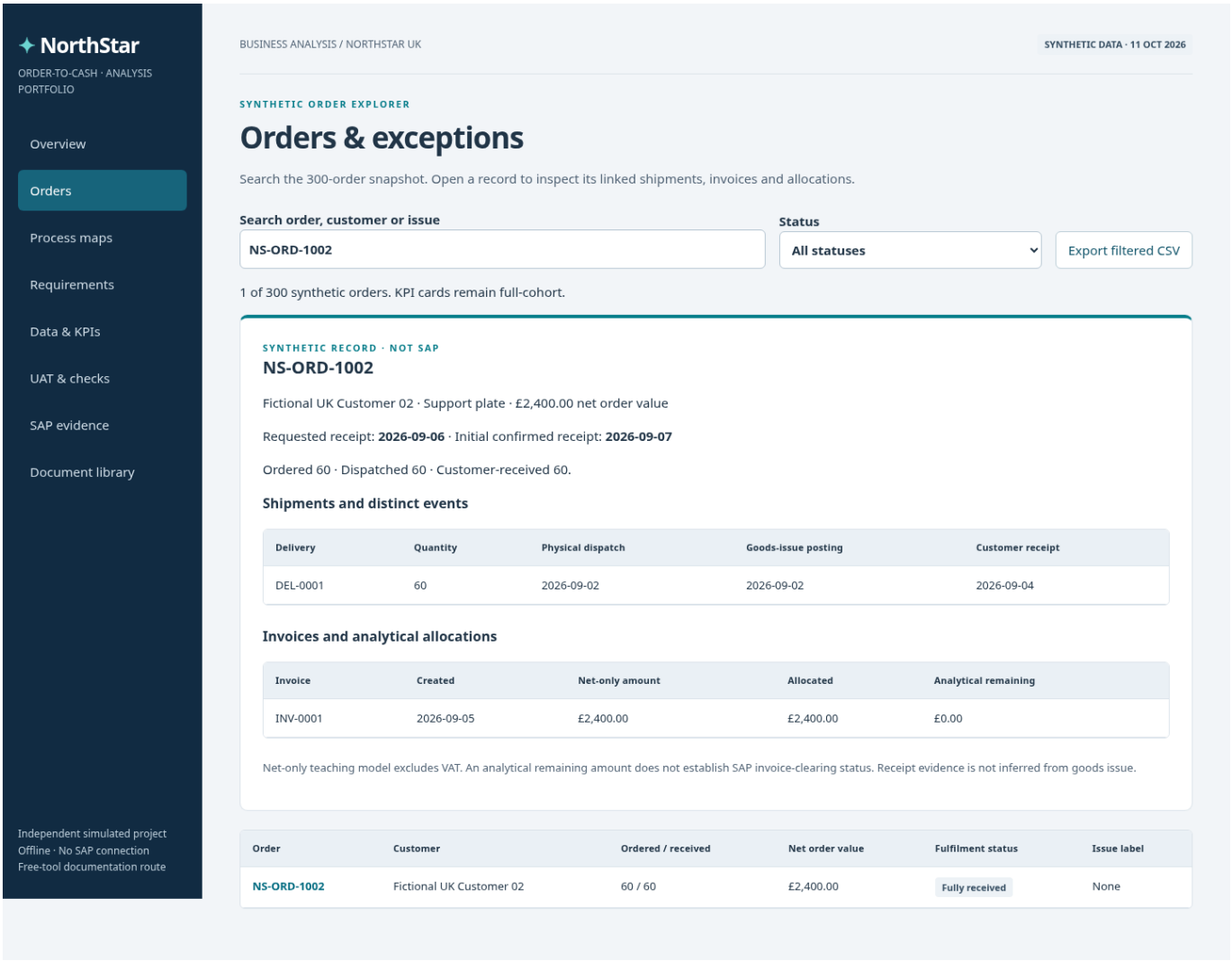
Full-cohort synthetic values and denominator limitations.



Actual local Chromium UI · screenshots/01-overview.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

Order drill-down

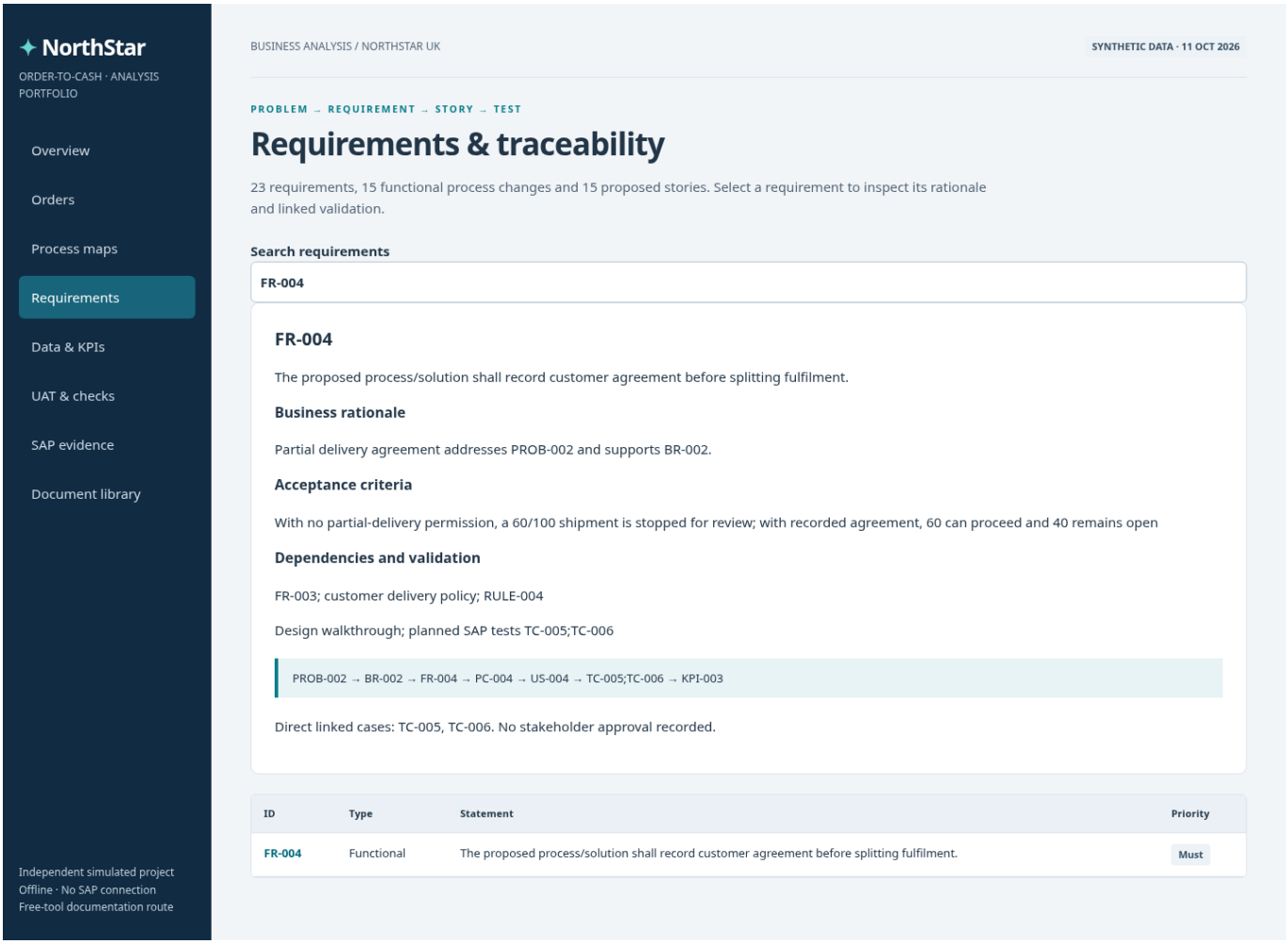
Actual synthetic NS-ORD-1002 with linked shipment, invoice and analytical payment amounts.



Actual local Chromium UI · screenshots/02-order-details.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

Requirements and traceability

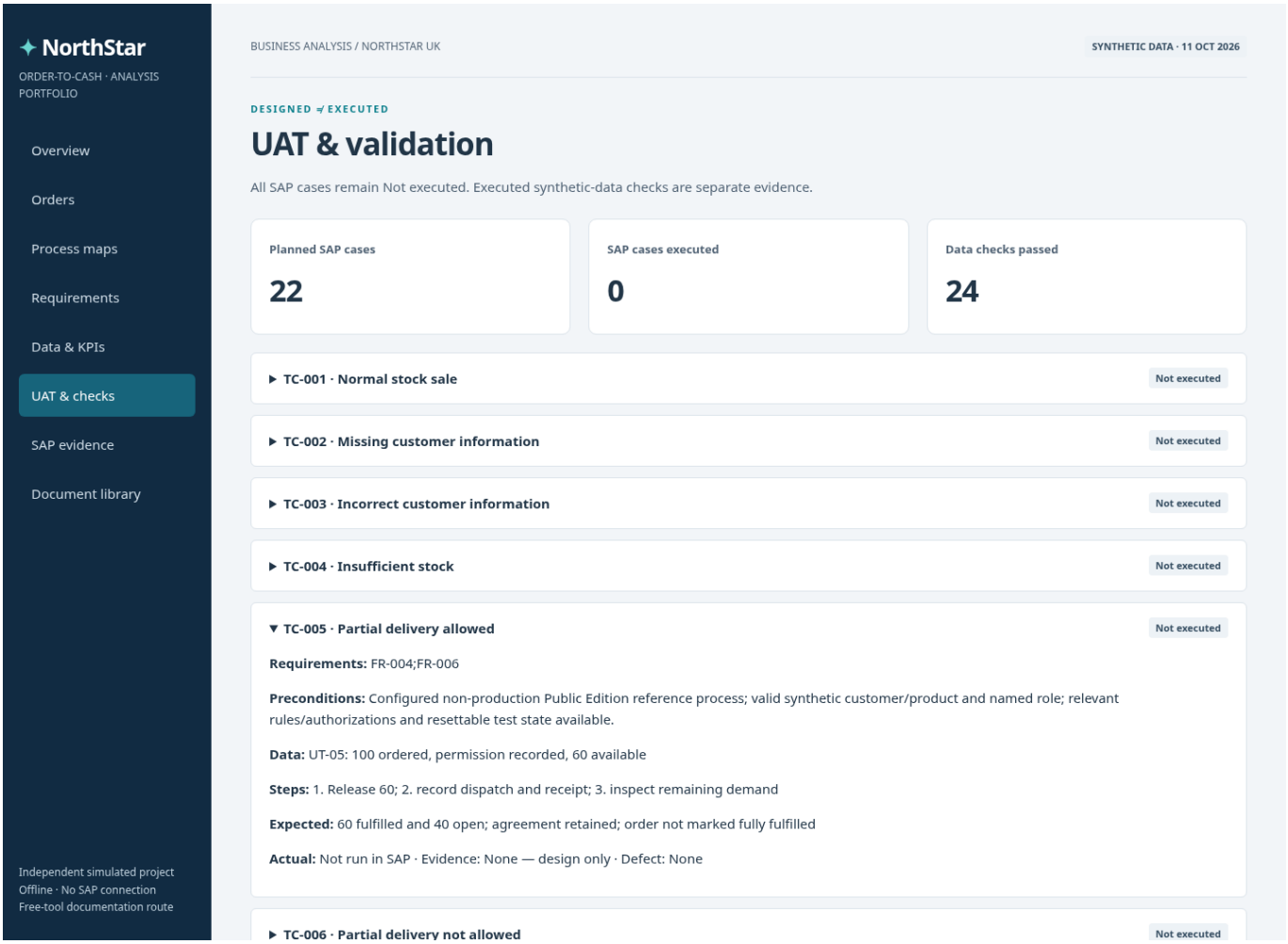
Partial-delivery requirement, acceptance criteria and linked evidence.



Actual local Chromium UI · screenshots/03-requirements.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

UAT design

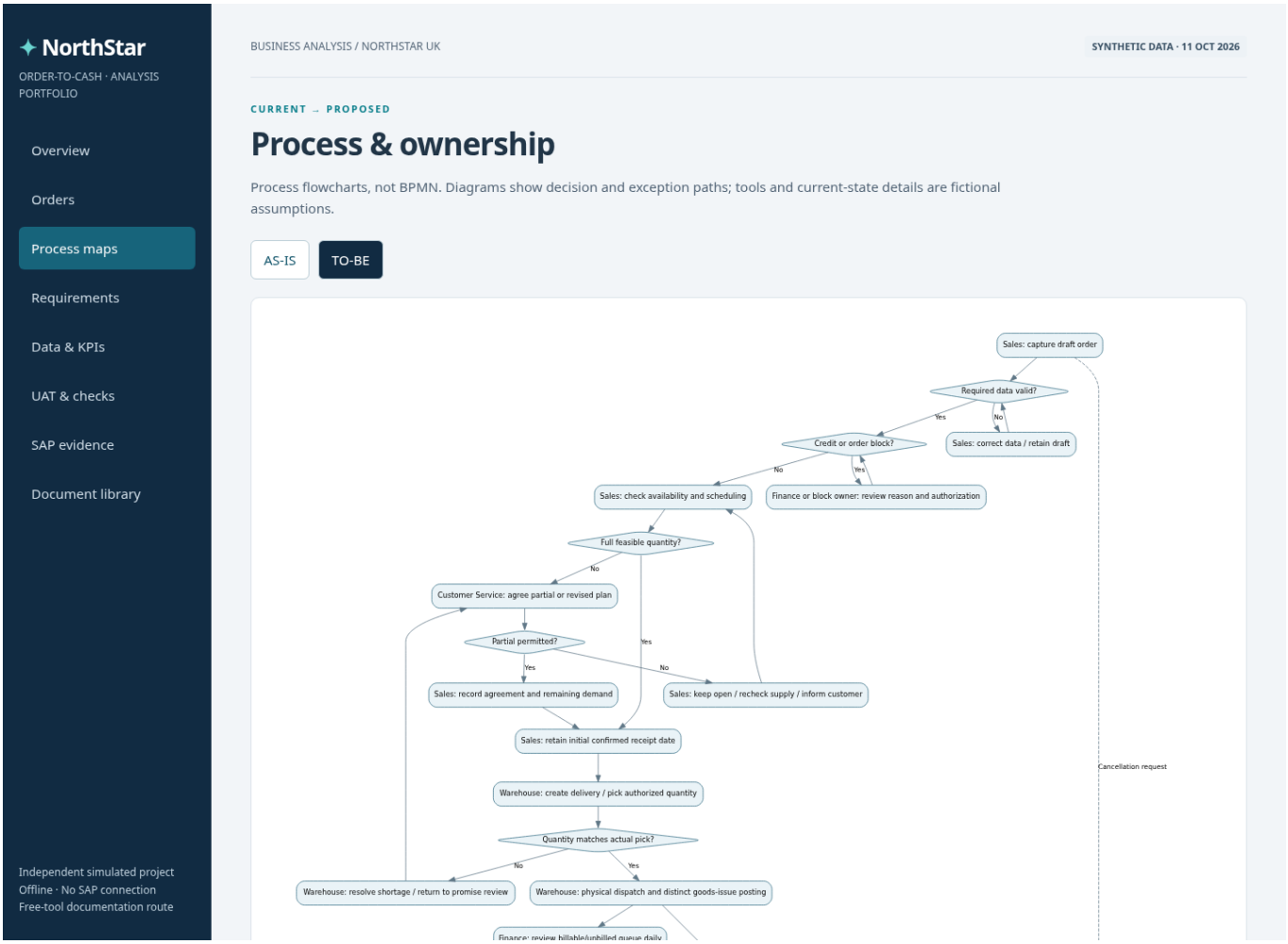
Test cases are readable; status remains Not executed in SAP.



Actual local Chromium UI · screenshots/04-uat.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

Proposed process flow

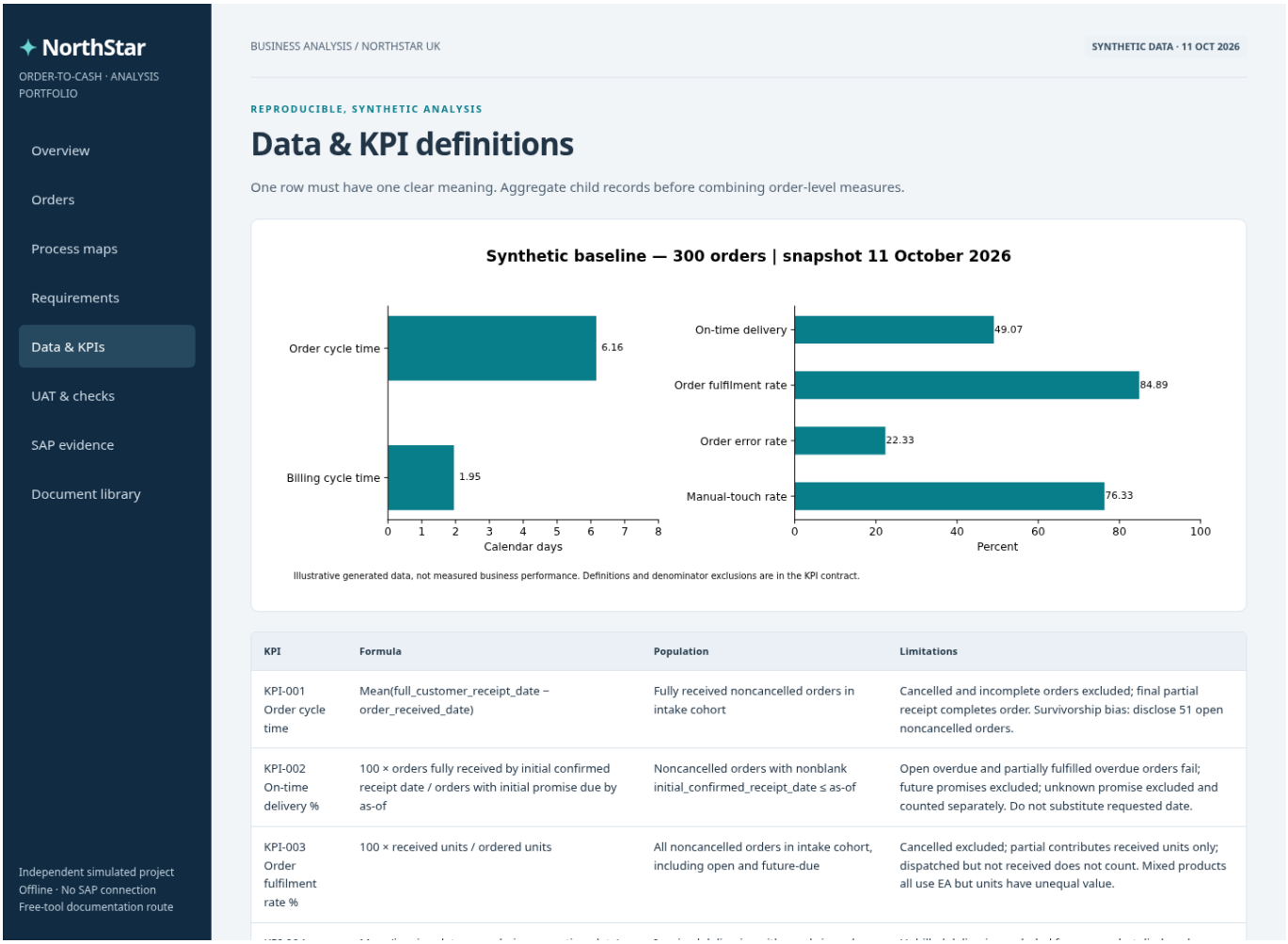
Actual process screen. This viewport shows the top of a scrollable full flowchart.



Actual local Chromium UI · screenshots/05-process.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

Data and KPI definitions

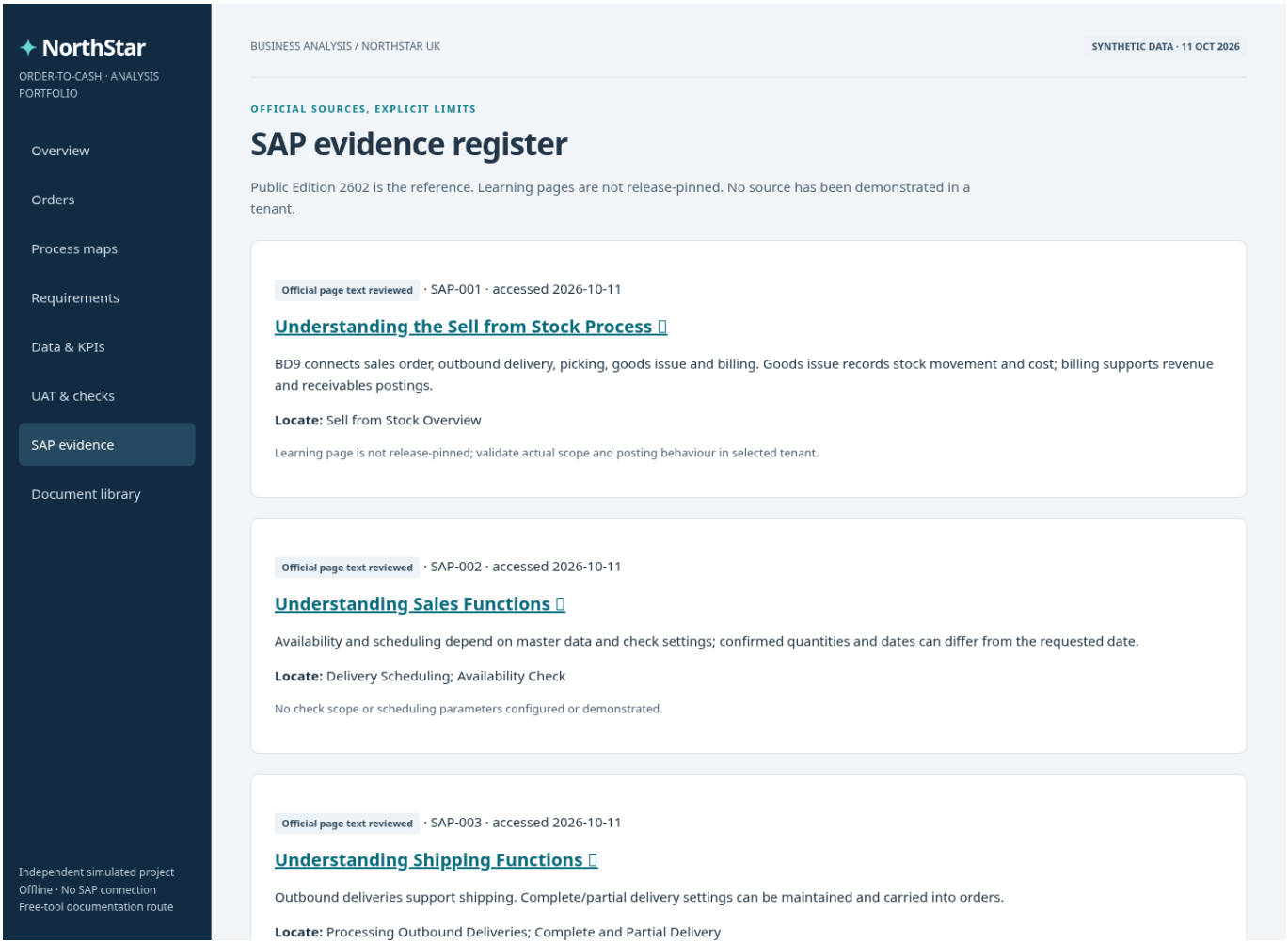
Separate day/percentage chart and the KPI calculation contract.



Actual local Chromium UI · screenshots/06-analysis.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

SAP evidence register

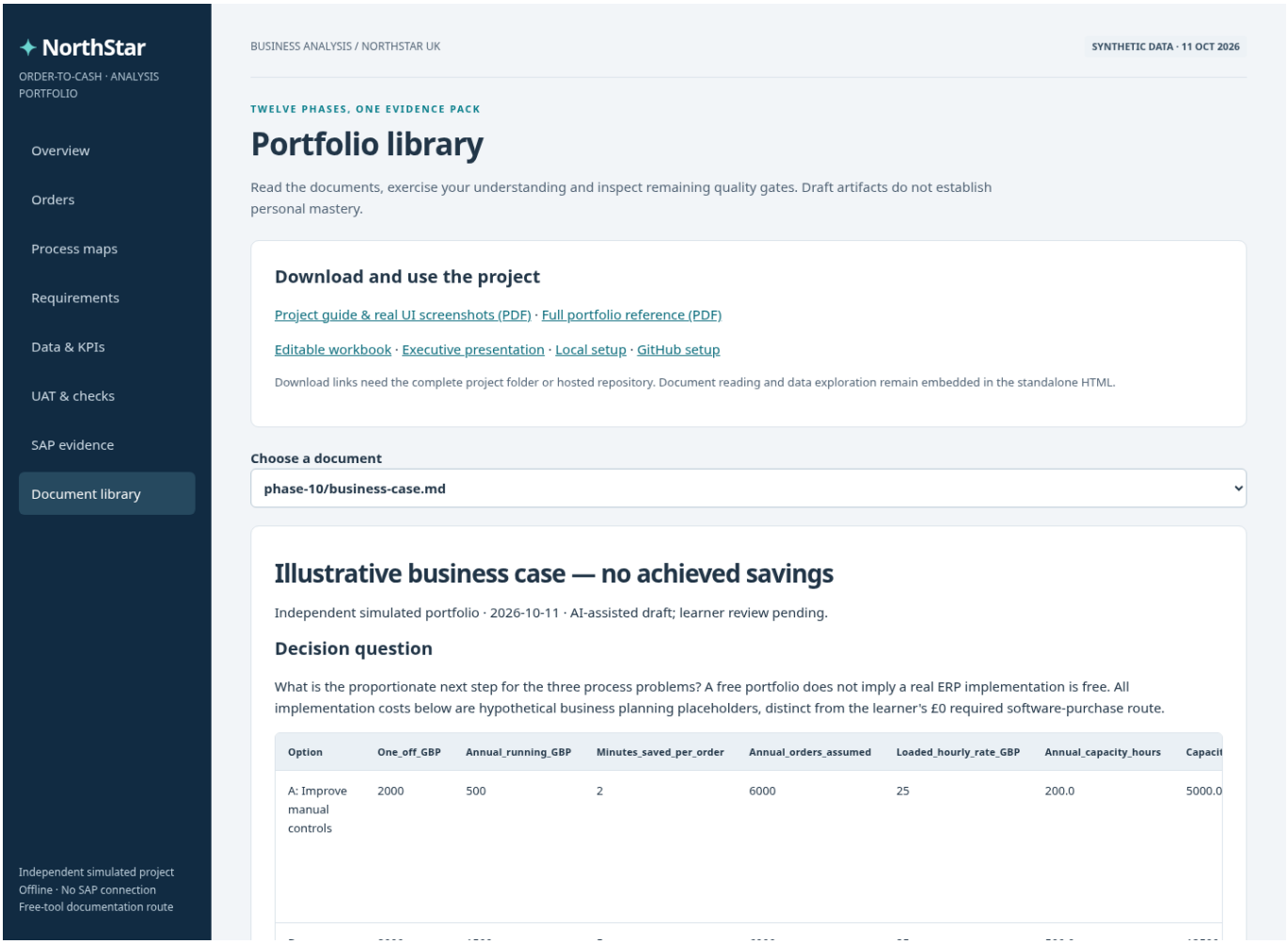
Reviewed documentation claims and scope limitations; no tenant execution.



Actual local Chromium UI · screenshots/07-evidence.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

Document library and downloads

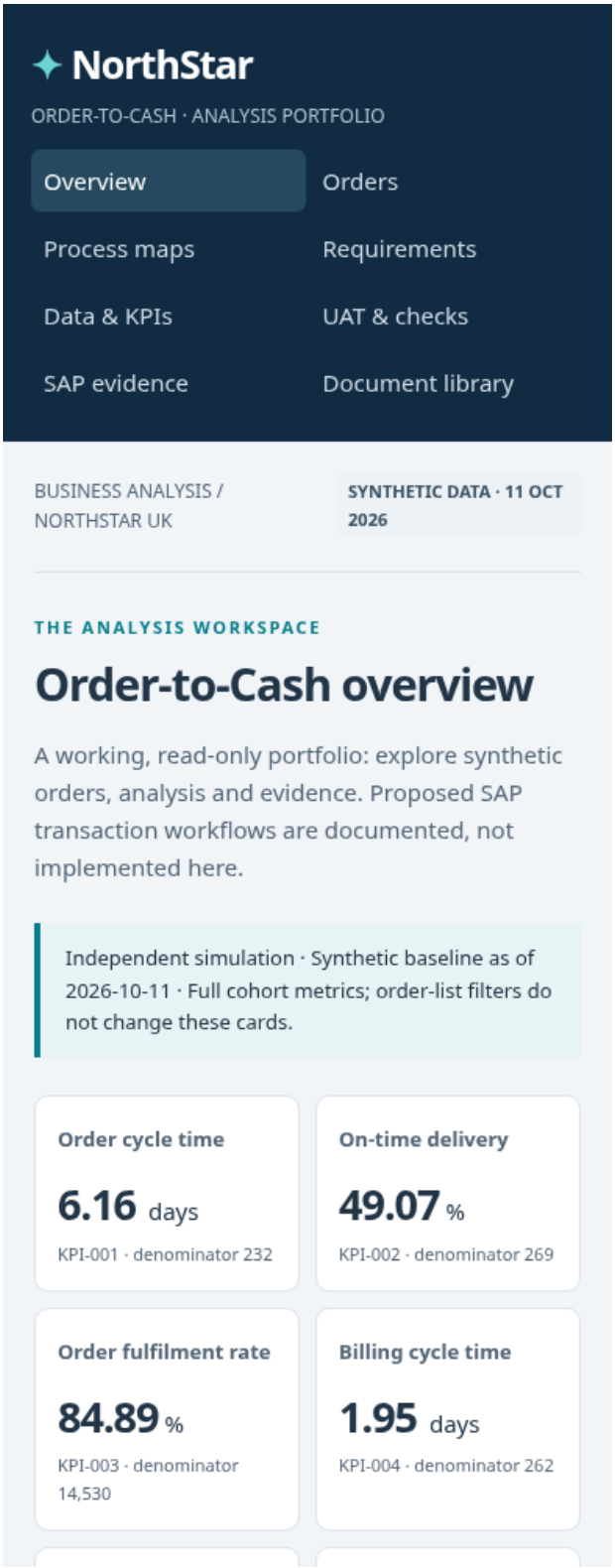
Choose phase documents and open the full repository download assets.



Actual local Chromium UI · screenshots/08-library.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

Mobile overview

Actual 390px Chromium viewport; not a physical-device test.



Actual local Chromium UI · screenshots/09-mobile.png · Read-only simulated portfolio, not SAP software. Open the PNG or zoom the PDF for fine text.

Run the project locally

The portfolio application is prebuilt. Viewing it does not require SAP, a database server, an account, npm packages or paid software. The editable workbook is a snapshot; it does not recalculate the browser data when changed.

Recommended: local web server

1. Extract the repository ZIP and open the `northstar-otc` folder. Keep the entire folder together.

2. Open a terminal inside that folder.

3. Run `python scripts/serve.py --open` (on Windows you may use `py scripts/serve.py --open`; on macOS/Linux, `python3 scripts/serve.py --open`).

4. If the browser does not open automatically, visit `http://127.0.0.1:8000/`.

5. Press Ctrl+C in the terminal when finished. If the port is occupied, run with `--port 8001`.

Python's standard library is enough for this server. Internet is unnecessary for the application itself; official SAP source links open external sites when selected. No records are written by the application. CSV export saves a file in your browser's download location.

Without Python

Double-click `index.html` in a modern browser. All interactive data and documents are embedded; no fetch calls or backend are required. Some managed browsers block file URLs. If that happens, use the local-server option or GitHub Pages. File-URL execution was blocked by this environment's browser policy; actual Chromium verification used HTTP on loopback. Do not interpret that policy restriction as a passed file-URL test.

Five-minute tour

1. Overview: read the six KPI cards and the denominator context.

2. Orders: search `NS-ORD-1001`; open it and notice that 60 on hand is not a confirmed promise. Filter other orders by fulfilment status and export the filtered CSV.

3. Requirements: search `FR-004`; inspect the story/test/KPI chain for partial delivery.

4. UAT: expand `TC-005` and `TC-006`; both remain Not executed in SAP.

5. Process maps: switch AS-IS and TO-BE.

6. Data & KPIs: inspect formulas and limitations; the cards remain full-cohort when the order list is filtered.

7. SAP evidence: distinguish reviewed sources from two unverified pages.

8. Document library: read each phase and download the PDF/workbook/presentation.

Rebuild instead of just viewing

Only contributors regenerating the data, charts and editable reports need the build tools. Install Python 3.11+ and Node.js 18+, then `python -m pip install -r requirements.txt` and `python scripts/build_all.py`. Graphviz is optional if the supplied rendered maps are retained. The Python data-only command is `python scripts/build_data.py`.

Optional real-browser checks and screenshot refresh require `python -m pip install -r requirements-browser.txt` plus an installed Chromium browser. Run `python scripts/browser_check.py`

`--browser-path /path/to/chromium`. That script creates a temporary local server, tests the app and closes both server and browser. It does not run SAP UAT. Windows Chrome is usually inside Program Files; provide its executable path if using it. Browser automation is optional for viewing or GitHub hosting.

After capturing screenshots, `python scripts/build_handover.py` regenerates the explanatory and full-reference PDFs. Run `python scripts/package_release.py` last to update the ZIPs. Regeneration overwrites generated files; reconcile direct Word/Excel edits with source Markdown/CSV/Python first.

Troubleshooting

- A blank page: use a current browser with JavaScript enabled; try the local server. Inspect the browser console and compare quality/browser-checks.json.

- Opening README shows text, not the app: open root `index.html` or the local URL.

- PDF link unavailable: confirm the full repository's downloads folder was extracted. A lone HTML copy keeps the app data but cannot carry separate downloadable files.

- Data edits do not appear: rebuild the viewer; it contains a generated snapshot rather than reading live CSVs.

- Changes disappear on rebuild: edit the source scripts, Markdown or CSV generation inputs, not only generated HTML/Office copies.

Push to GitHub and open the website

GitHub stores the whole repository. GitHub Pages serves its static browser application. Pushing files alone does not enable Pages; the publishing source must be selected once.

Free route

Use a **public repository** with GitHub Free and the default github.io address. No custom domain, SAP subscription or paid backend is needed. The repository contains synthetic records and simulated analysis. The site and source will be public, so add only information you intend to share. The chat/cloud workspace's own pricing is separate.

Option A — Git commands

Create a new empty public repository named `northstar-otc` on GitHub. Extract this project's ZIP and open a terminal inside the folder containing index.html, README.md, viewer and scripts. Replace YOUR_USERNAME in the remote URL below.

```
git init
git add .
git commit -m "Add simulated NorthStar OTC portfolio"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/northstar-otc.git
git push -u origin main
```

Authenticate using your normal GitHub/Git credential workflow. Do not put tokens in source files. If a remote already exists, inspect it with `git remote -v` before changing anything. The downloaded folder contains no .git history, credentials or configured remote.

Option B — GitHub Desktop

Create a repository from the extracted project folder in GitHub Desktop, make the initial commit and choose Publish repository. Select public visibility for the GitHub Free Pages route. Ensure `index.html` is at the repository root, not inside an extra nested northstar-otc folder.

Enable Pages once

In the repository, open Settings → Pages. Under Build and deployment, choose **Deploy from a branch**, select **main** and **/(root)**, then Save. Wait for GitHub's deployment result. The usual project URL is `https://YOUR_USERNAME.github.io/northstar-otc/`; copy the actual URL shown by GitHub. Subsequent pushes to the selected branch update the site.

The root index.html is self-contained. Relative file links work under a repository subpath. The supplied .nojekyll prevents Jekyll processing; no Node/Python build is needed to host the prebuilt application. No GitHub Actions deployment workflow is required for this route. The repository ignores generated ZIP archives to avoid recursively committing release bundles; PDFs, screenshots, code, data and editable reports remain included.

Check your deployed site

Open the actual Pages URL, navigate all sections, search NS-ORD-1001, export a CSV, open a PDF and test on a phone. GitHub deployment has not been run or verified by this handover. Local testing does not claim a live URL exists.

Sources checked 11 October 2026

- GitHub Pages availability and static hosting: https://docs.github.com/en/pages/getting-started-with-github-pages/what-is-github-pages

- Publishing from a branch and root folder: https://docs.github.com/en/pages/getting-started-with-github-pages/configuring-a-publishing-source-for-your-github-pages-site

The first source confirms GitHub Free supports Pages for public repositories. The second describes the branch/folder configuration above. Future GitHub product or account changes should be checked against the official documentation.